

june 2026



C-SECTION POSTPARTUM PORTAL TRIAGE USING NLP

Introduction to Natural Language Processing

**A Deep Learning Framework for Clinical Severity Triage
of Post-C-Section Patient Messaging**

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CLINICAL BACKGROUND & PATIENT ENGAGEMENT

Post-C-Section Care & The "Fourth Trimester"

The Critical 42 Days

High-risk postpartum window for dynamic complications

ACOG Global Guideline

Recommends replacing the single 6-week visit with ongoing communication via patient portals.

Clinical Guidance (Israel)

Providers (Clalit, Sheba..) supply strict protocols for active at-home monitoring of maternal "Red Flags".

Patient Preference

High shift toward digital reporting over physical visits due to post-surgical pain & newborn care.

The Data Challenge

Incoming clinical messages arrive as unstructured, noisy, conversational free-text.

THE INBOX CRISIS & THE NEED FOR AI

The Inbasket Burden

Surging Volume

Asynchronous portal messages increased by **157%**.

The Mix

Merges routine administrative requests with acute patient free-text

pandemic has transformed how patients expect to interact with their providers, accelerating both virtual care and messaging. In a study of 245 US health systems using Epic EHR, patient *In Basket* messages increased by 157% compared to pre-pandemic levels.⁴ Another study found a doubling in the number of patient messages received per day, jumping at the onset of the pandemic and sustaining.¹ With this increasing

The FI-FO Danger

Chronological Review

Messages are read manually based on standard arrival order

Delayed Care

Critical clinical warnings wait in queue behind non-urgent update



שליחת בקשה לרופא/ה בנושאים:

חידוש מרשם, אישור מחלה, שאלה לרופא/ה, סיכום מצב רפואי והפניה

The AI Triage Solution

Advanced Transformers

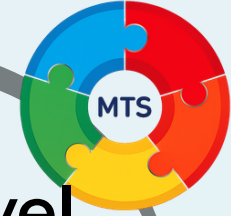
Decoding informal slang using BERT and advanced models

Benchmarking

Evaluating deep architectures to find the optimal clinical encoder

MTS Mapping

Classifying free-text into the 5 Manchester Triage priority levels



Adapted the 5 level MTS to a tailored **4 level triage** system for postpartum home care

PROJECT TASK DESCRIPTION

BASIC STRUCTURE

Prediction Task

Binary classification of routine (0) and emergency (1) + segmentation of Acute Urgency (1/2/3).

System Pipeline Inputs

Unstructured, noisy free-text portal messages from patients and caregivers.

Project Novelty

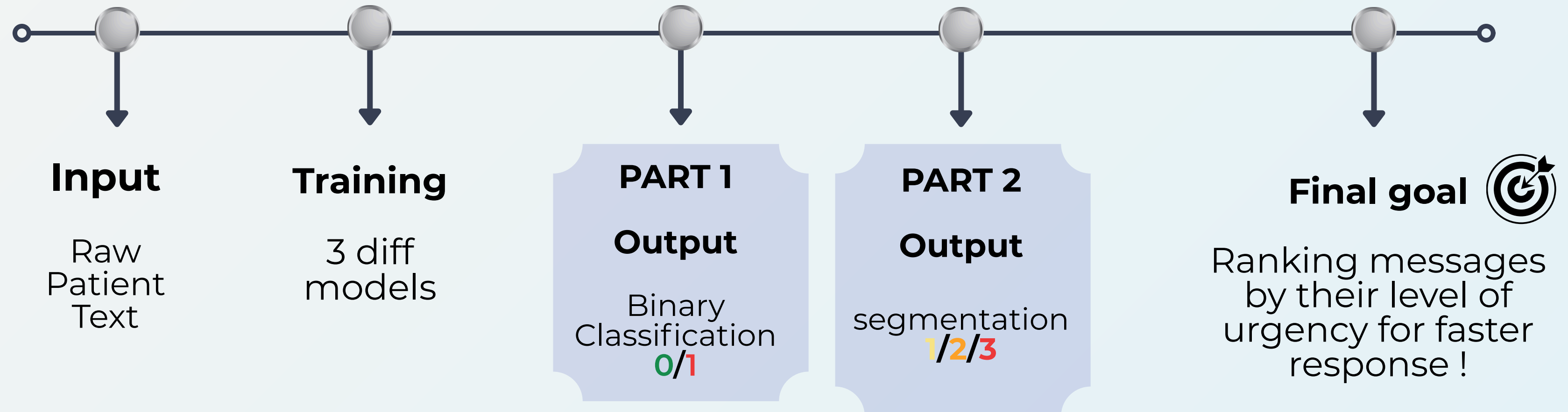
- Processes raw consumer-voiced language in Several variations.
- Enforces hard logic constraints to link global alerts directly to active triggers.

Level 0 (Routine)
Admin requests
(refills, sick notes)
or standard healing.

Level 1 (Low)
Mild localized pain,
dysuria, or minor
incision leakage

Level 2 (Moderate)
Spreading redness,
purulent drainage, or
persistent fever

Level 3 (Immediate)
Postpartum
hemorrhage, sepsis
signs, or suspected
DVT/PE.



PROJECT TASK DESCRIPTION

DATA, MODELS & METHODS

Data focus & Generation Planning

– Medical data is often confidential to the general public –

Artificial data

Generating a custom synthetic dataset based on prompts and examples

Target Columns (Schema)

Patient_Message: Raw conversational text inputs
Urgency_Level: Multi-class target label (0 to 3) based on clinical severity

Data Format Structure

Standardized into structured JSON payloads

Models and Methods

Model 1 – BIO BERT

Fine-tuned and pre-trained

Model 2 – Bi-LSTM

Serial and bidirectional neural network

Model 3 – Zero-Shot GPT-4

Generative LLM via Prompting Evaluated directly without weight adjustments or training

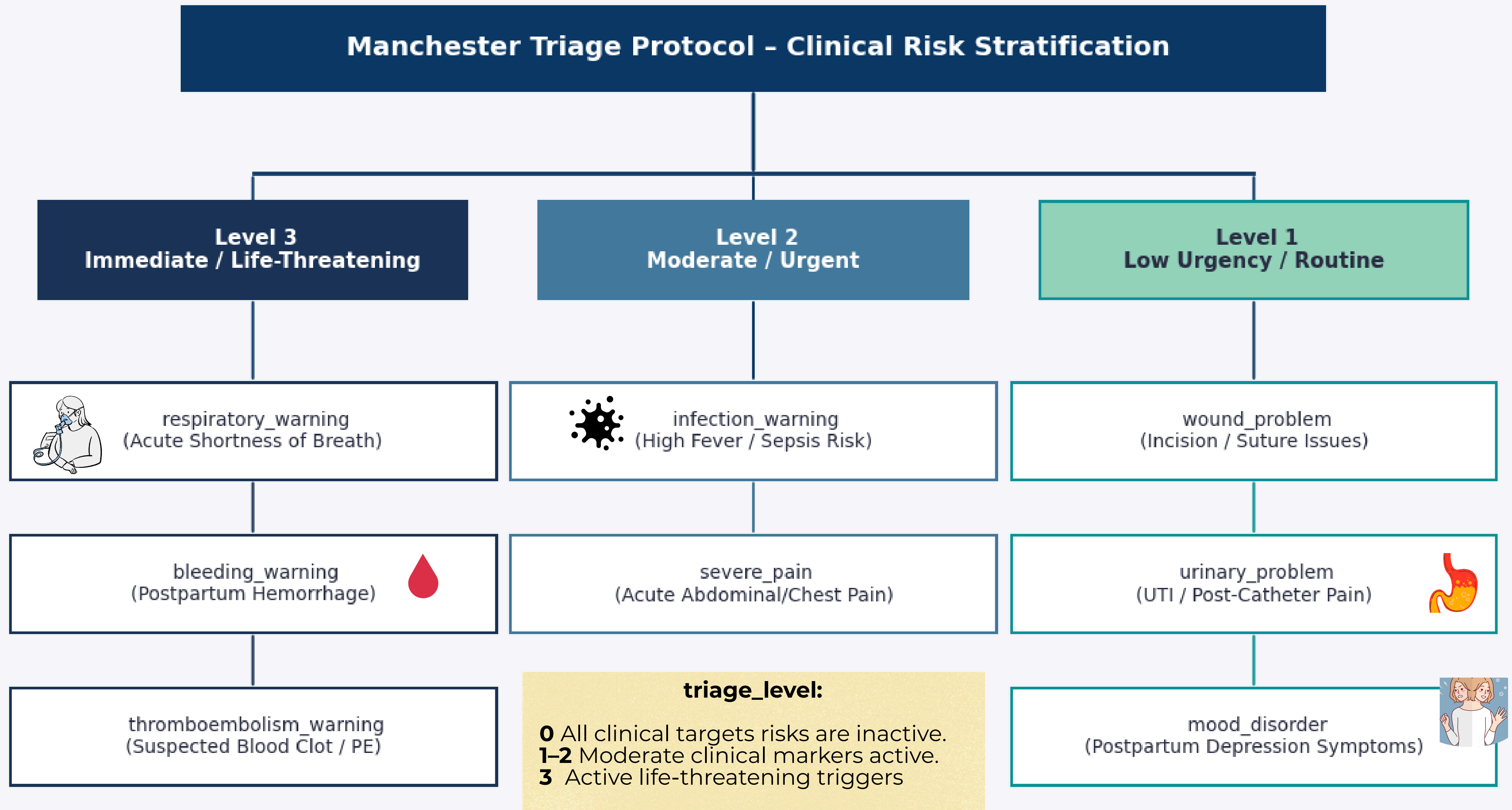
Comparison of methods

contextual understanding **vs.** Evaluate Sequential Learning **vs.** Prompt-based LLM

DATASET & DATA GENERATION

Dataset Overview	Generation Methodology	Labeling Pipeline
Target Population Post-C-section patients tracking their 42-day recovery window Data Nature Synthetic conversational text mimicking patient-clinician messaging. Volume & Structure 3,000 distinct clinical reports	Engine Generated using advanced Language Models (OpenAI API) Prompt Engineering Designed with highly specific personas, clinical constraints, and linguistic to ensure realistic patient phrasing. Medical Taxonomy Structured around 8 critical post-C-section warning signs	clinical categories Labeling We have extracted 8 common emergencies (based on articles) which we have categorized into 1-3 levels of emergency. If one of the categories appears, its column is marked 1, and accordingly the emergency column 1-3 Traige label final multi-class urgency label (0-3)

TARGET CLINICAL SCHEMA DEFINITIONS



Data Examples

Case Example (Triage Level 3)



case_id 5
patient_message "My partner just had a C-sec 2 days ago, and she's in really bad shape right now. She's complaining about this intense pain in her left leg, it's huge and feels hot and s
he says it's purple too. I'm so scared, she's pretty much soaked the pad already, like it's just pouring out, and there's these clots. I've been trying to get her to breathe, she started panic
king and feels like she can't get enough air. Plus, she's burning up, like over 39 degrees, and can't stop shivering. I know this is serious, we can't just wait anymore. Can you tell me what
to do, or should I just take her to the ER? Pls let me know ASAP. Thanks."
sender "caregiver"
length_target "long"
cat_wound_problem 0
cat_infection_warning 1
cat_bleeding_warning 0
cat_respiratory_warning 0
cat_severe_pain 1
cat_thromboembolism_warning 0
cat_mood_disorder 0
cat urinary problem 0
triage_level 3
char_count 625
word_count 120
evidence "soaked the pad already, like it's just pouring out, and there's these clots"

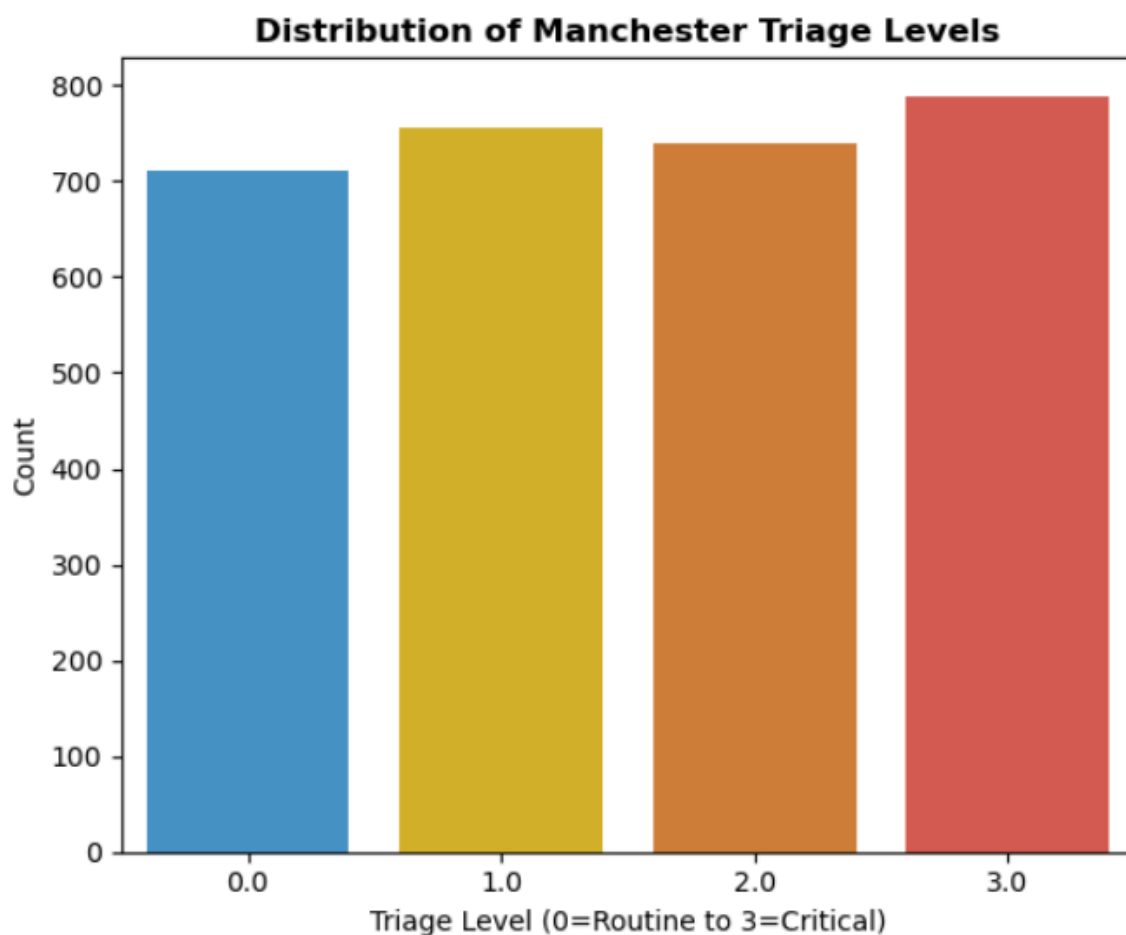
case_id 3
patient_message "Hey there, so my partner just had her C-sec a couple of days ago and she's been feeling super tired but the doc said that's normal? She says the soreness is mild but
definitely there, like, she's having this light pinkish discharge w/ no big clots or anything, it weirds me out just a bit. Should I worry about that?? I don't think it's anything serious thoug
h, I read about post-op stuff online, but it just seems a bit much for her to handle at once, you know? Her OBGYN said recovery would take a few wks, but right now, benadryl isn't cutti
ng the sleeplessness... just asking bc idk what's typical?? TMI but I saw her covering a little drop here and there every now and then... is all that ok though? Msg back when you can, tha
nks!"
sender "caregiver"
length_target "long"
cat_wound_problem 0
cat_infection_warning 0
cat_bleeding_warning 0
cat_respiratory_warning 0
cat_severe_pain 0
cat_thromboembolism_warning 0
cat_mood_disorder 0
cat urinary problem 0
triage_level 0
char_count 735
word_count 138
evidence ""

Case Example (Triage Level 0)

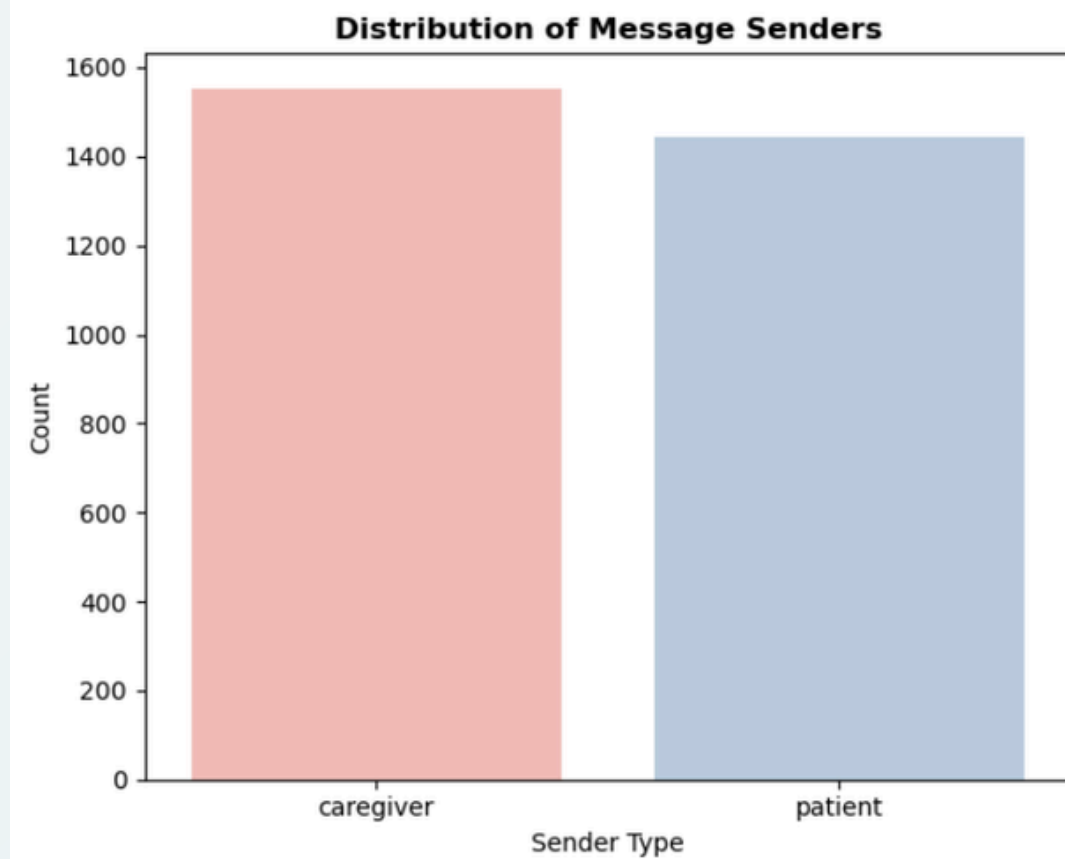


EDA

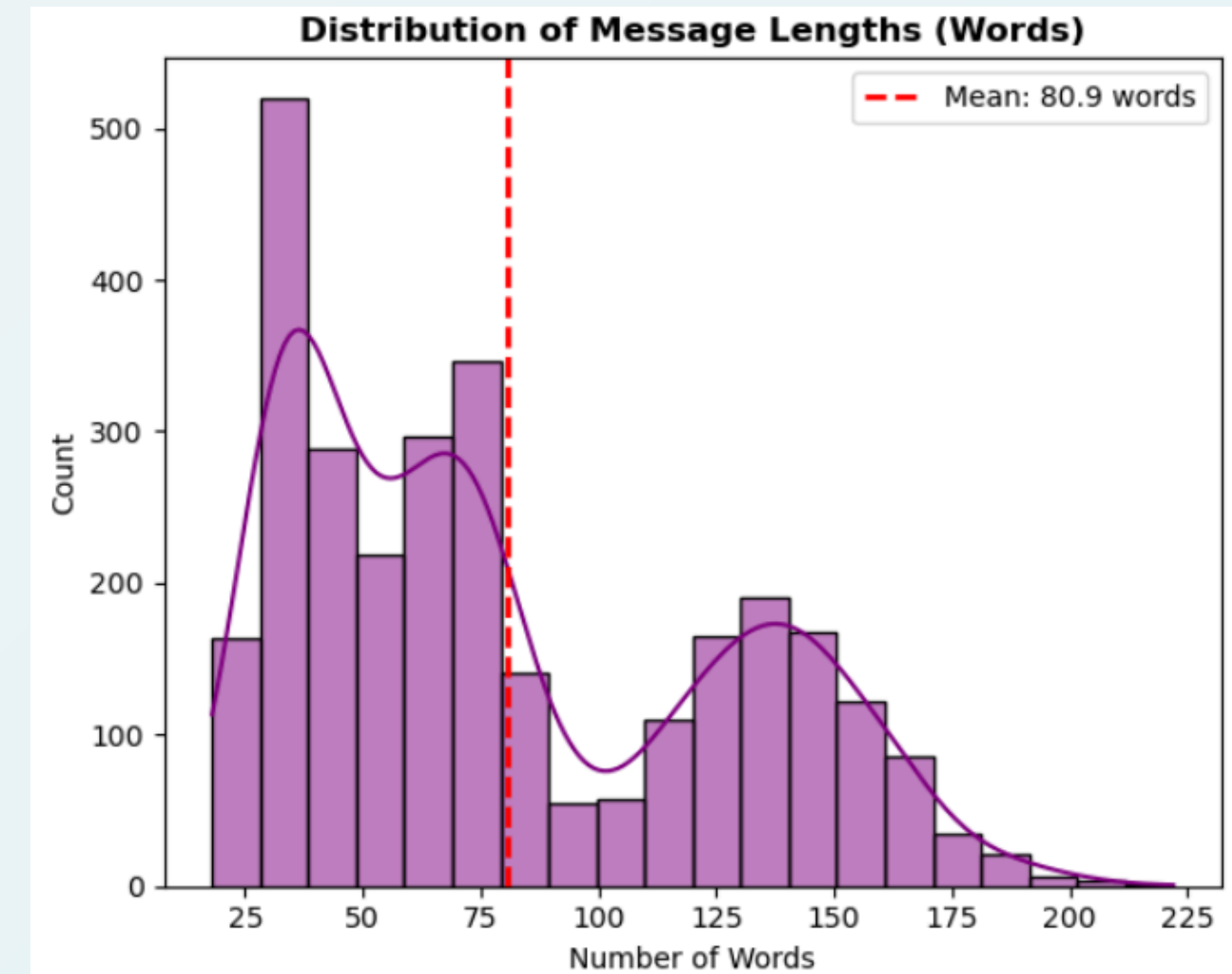
```
=== 📄 Manchester Triage Level Distribution ===  
Level 0.0: 712 samples (23.7%)  
Level 1.0: 755 samples (25.2%)  
Level 2.0: 740 samples (24.7%)  
Level 3.0: 789 samples (26.3%)  
Level 347.0: 1 samples (0.0%)
```



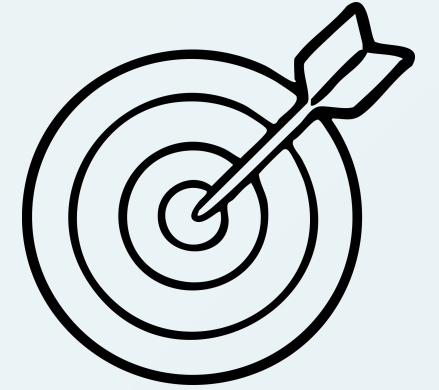
```
=== 👤 Sender Distribution ===  
Caregiver: 1553 samples  
Patient: 1442 samples
```



```
=== 📏 Message Length Statistics (in Words) ===  
Average words per message: 80.9  
Max words in a message: 222.0  
Min words in a message: 0.0
```



PROJECT PERFORMANCE METRICS & KPIS



Macro F1-Score:

Our primary KPI to evaluate performance fairly across all 4 triage levels.

Overall Accuracy:

Measuring the total percentage of correct triage predictions across the entire dataset

Cross-Model Benchmark:

Provides a standard baseline to visually compare the raw predictive power of Bi-LSTM, GPT, and ClinicalBERT

Dataset Splitting & Validation

Stratified 60/15/25 Split:

Data is partitioned into
Train (60%),
Validation (15%),
and Test (25%) sets.

Proportional Preservation:

Splitting is stratified based
on triage_level to maintain
identical class distributions
across all subsets.

Loss Tracking:

Stage 1 (Binary Filter)
Optimized via **BCE Loss**
(Routine vs. Urgent).

Stage 2 (Severity Regressor)
Optimized via **MSE Loss**
(Minimizing clinical distance).

Overfitting Control
Monitored via Validation Loss +
Early Stopping.

BASELINE - MODEL (BIO BERT) PART 1

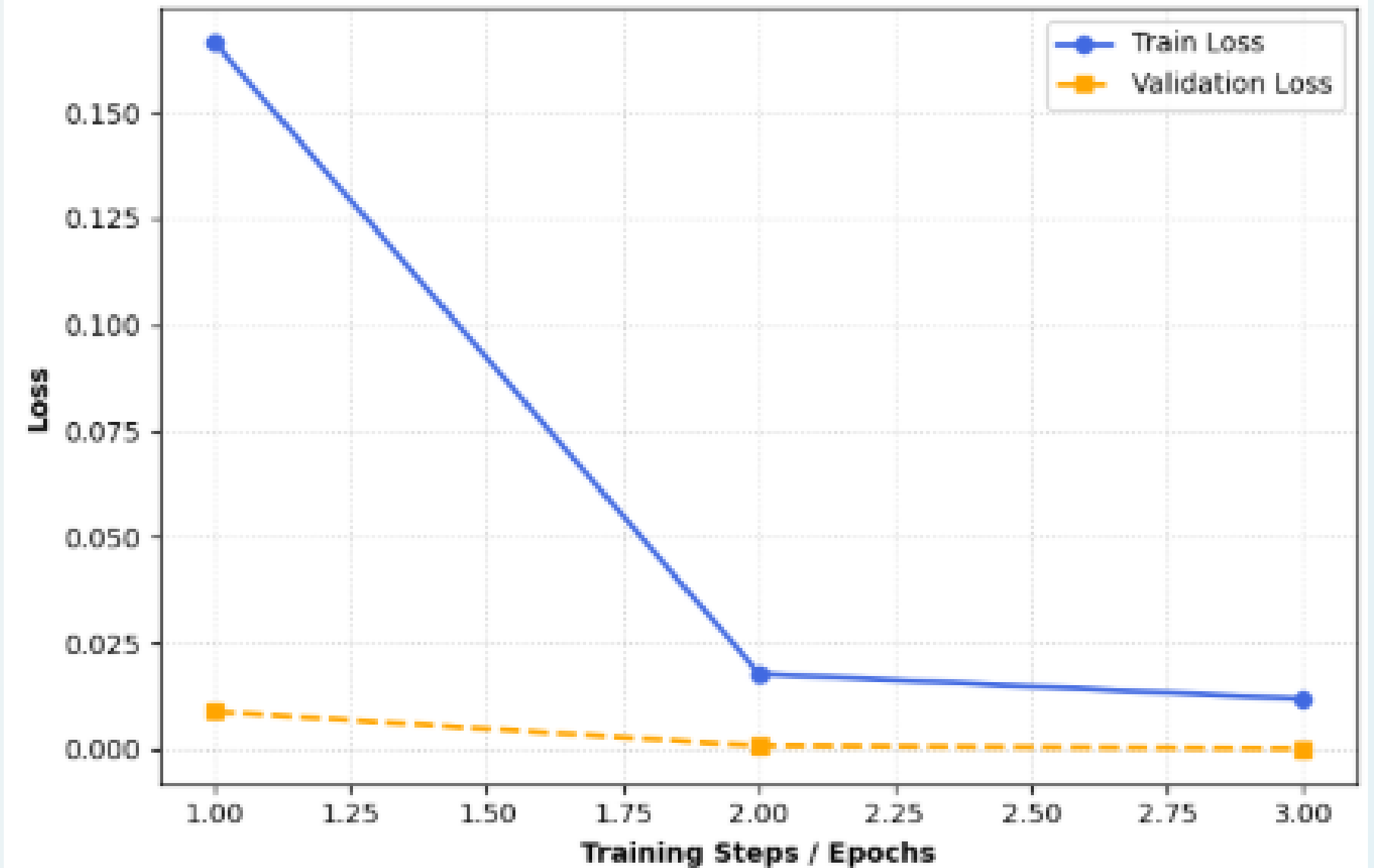
• STAGE 1 (Binary Filter - Routine vs Urgent):

- Accuracy: 0.9787
- Macro F1-Score: 0.9697

📄 Detailed Stage 1 Report:

	precision	recall	f1-score	support
Routine (0)	1.00	0.91	0.95	179
Urgent (1)	0.97	1.00	0.99	571
accuracy			0.98	750
macro avg	0.99	0.96	0.97	750
weighted avg	0.98	0.98	0.98	750

Stage 1: Binary Filter Loss (Cross-Entropy)



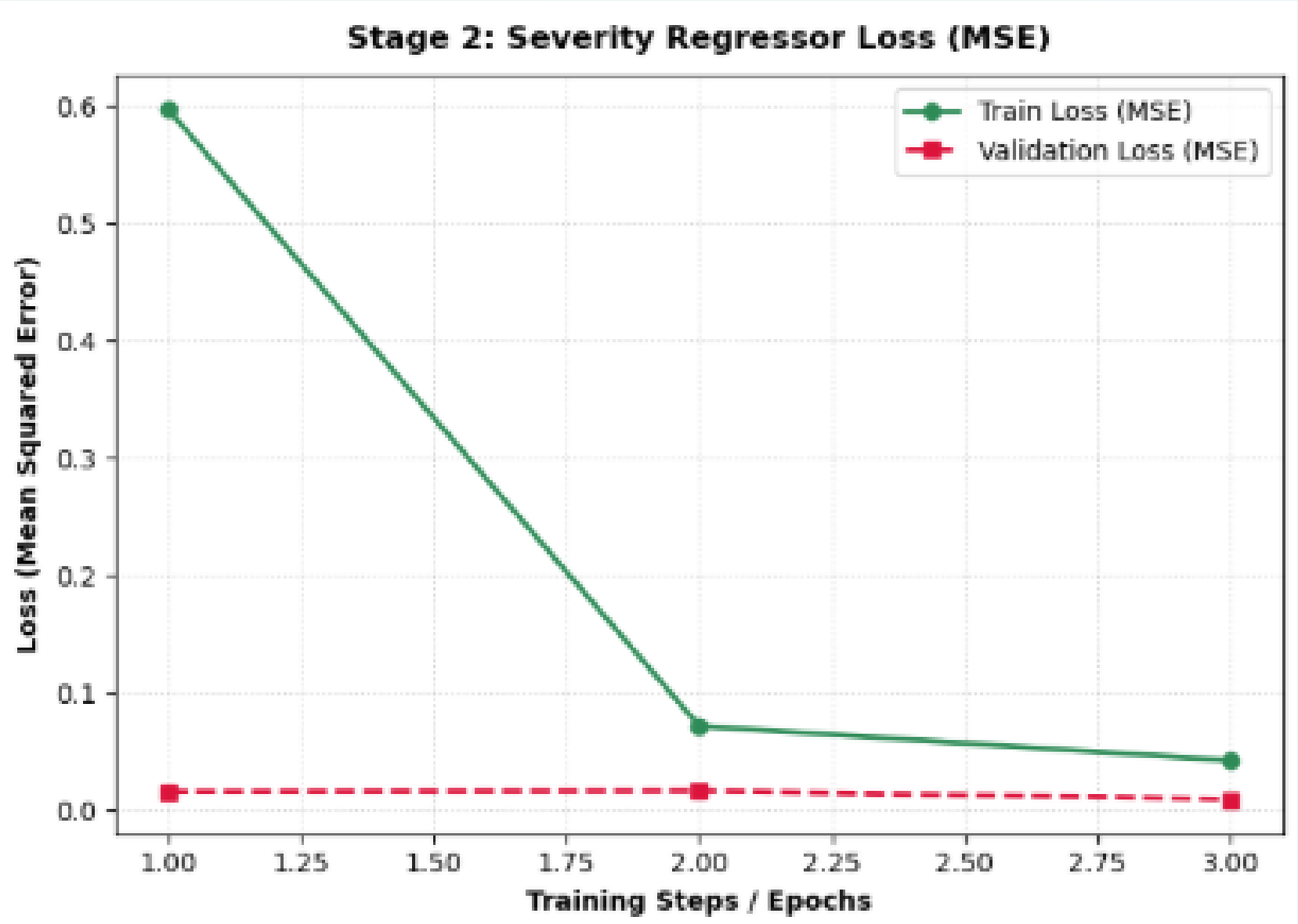
BASELINE - MODEL (BIO BERT) PART 2

• STAGE 2 (Severity Regressor - Levels 1-3 Only):

- Mean Squared Error (MSE): 0.0092
- Accuracy (Rounded): 0.9947
- Macro F1-Score (Rounded): 0.9947

📄 Detailed Stage 2 Report:

	precision	recall	f1-score	support
Level 1	1.00	0.99	1.00	189
Level 2	0.98	1.00	0.99	185
Level 3	1.00	0.99	0.99	197
accuracy			0.99	571
macro avg	0.99	0.99	0.99	571
weighted avg	0.99	0.99	0.99	571



BASELINE - MODEL (BIO BERT) OVERALL

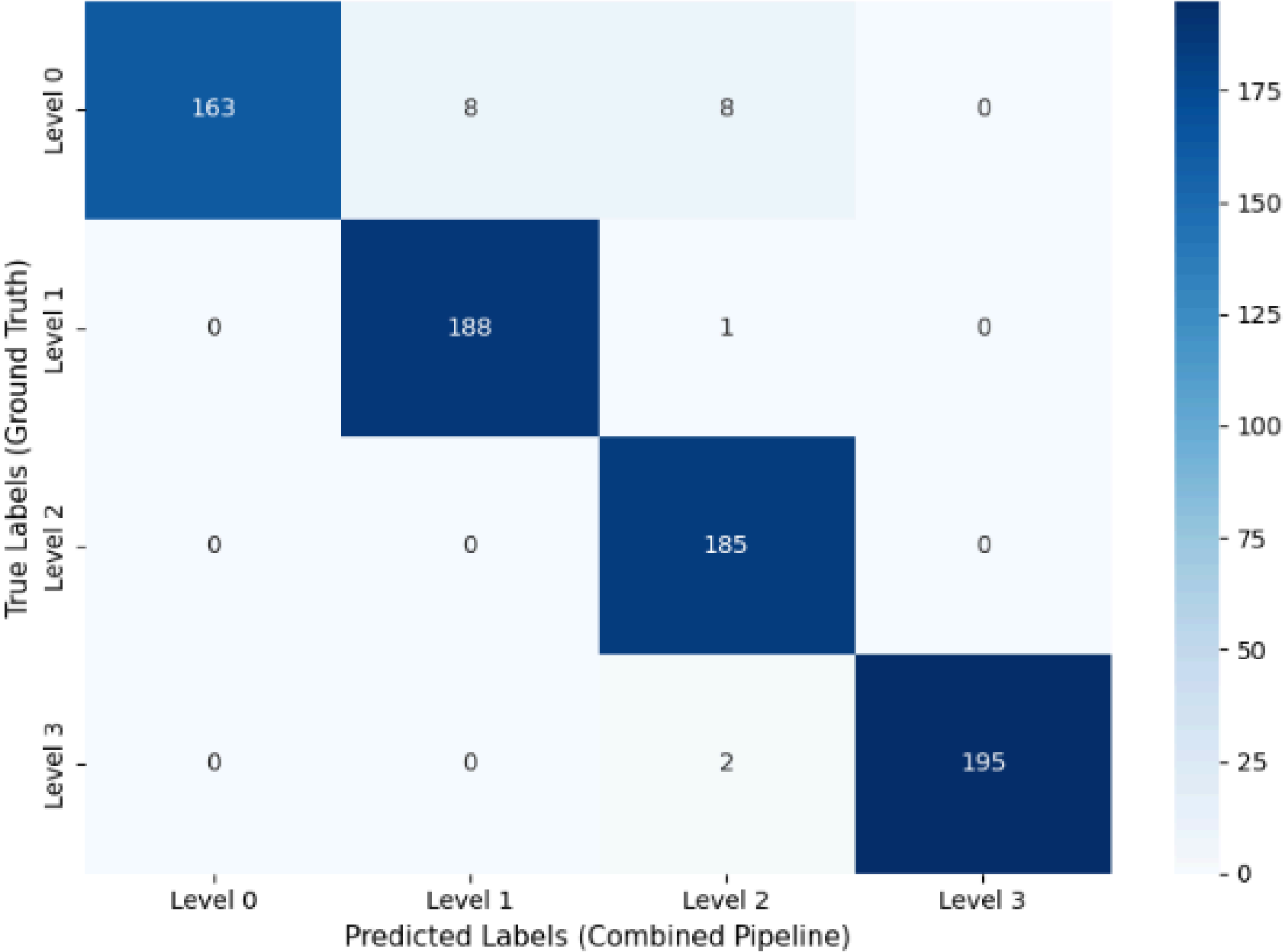
👑 FINAL COMBINED CASCADE PIPELINE (Full Model 0-3):

- Global Accuracy: 0.9747
- Global Macro F1-Score: 0.9740

📄 Final Full Pipeline Classification Report:

	precision	recall	f1-score	support
Level 0	1.00	0.91	0.95	179
Level 1	0.96	0.99	0.98	189
Level 2	0.94	1.00	0.97	185
Level 3	1.00	0.99	0.99	197
accuracy			0.97	750
macro avg	0.98	0.97	0.97	750
weighted avg	0.98	0.97	0.97	750

BioBERT Cascade Pipeline (Sigmoid + MSE) Confusion Matrix



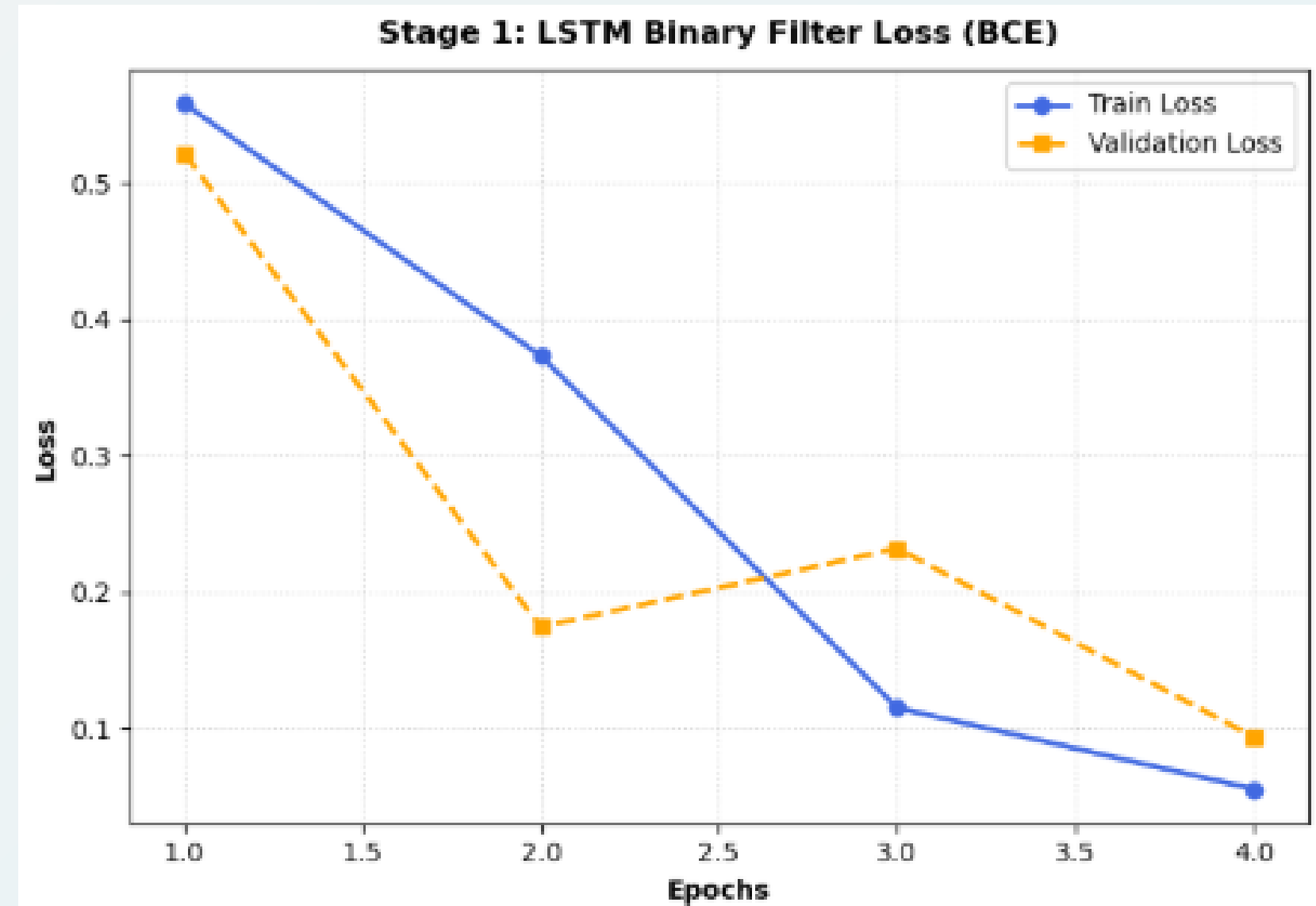
COMPLEX - BI LSTM PART 1

★ STAGE 1 (Binary Filter - Routine vs Urgent):

- Accuracy: 0.9813
- Macro F1-Score: 0.9737

📄 Detailed Stage 1 Report:

	precision	recall	f1-score	support
Routine (0)	0.99	0.93	0.96	179
Urgent (1)	0.98	1.00	0.99	571
accuracy			0.98	750
macro avg	0.99	0.96	0.97	750
weighted avg	0.98	0.98	0.98	750



COMPLEX - BI LSTM PART 2

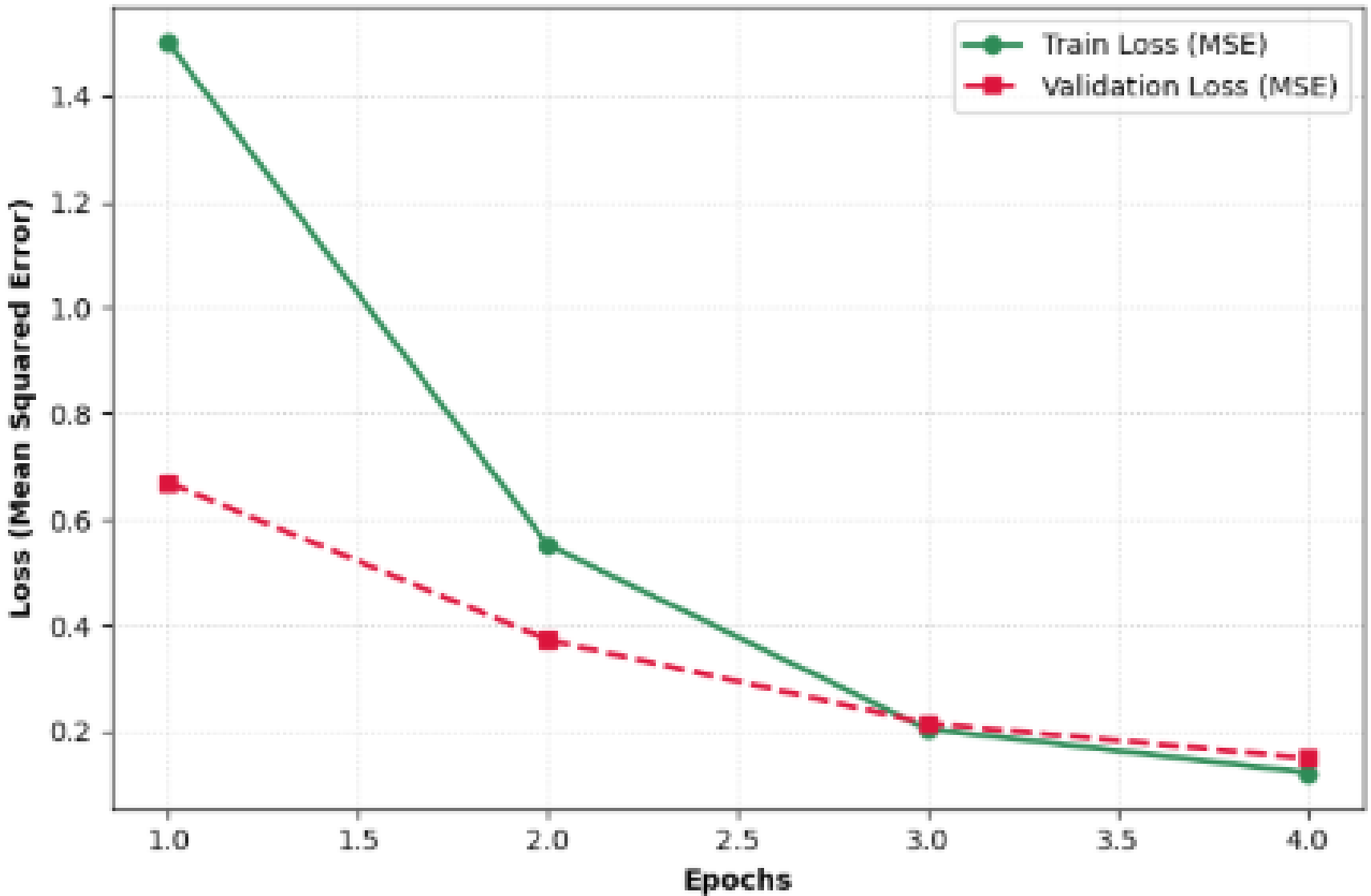
• STAGE 2 (Severity Regressor - Levels 1-3 Only):

- Mean Squared Error (MSE): 0.1579
- Accuracy (Rounded): 0.8074
- Macro F1-Score (Rounded): 0.8070

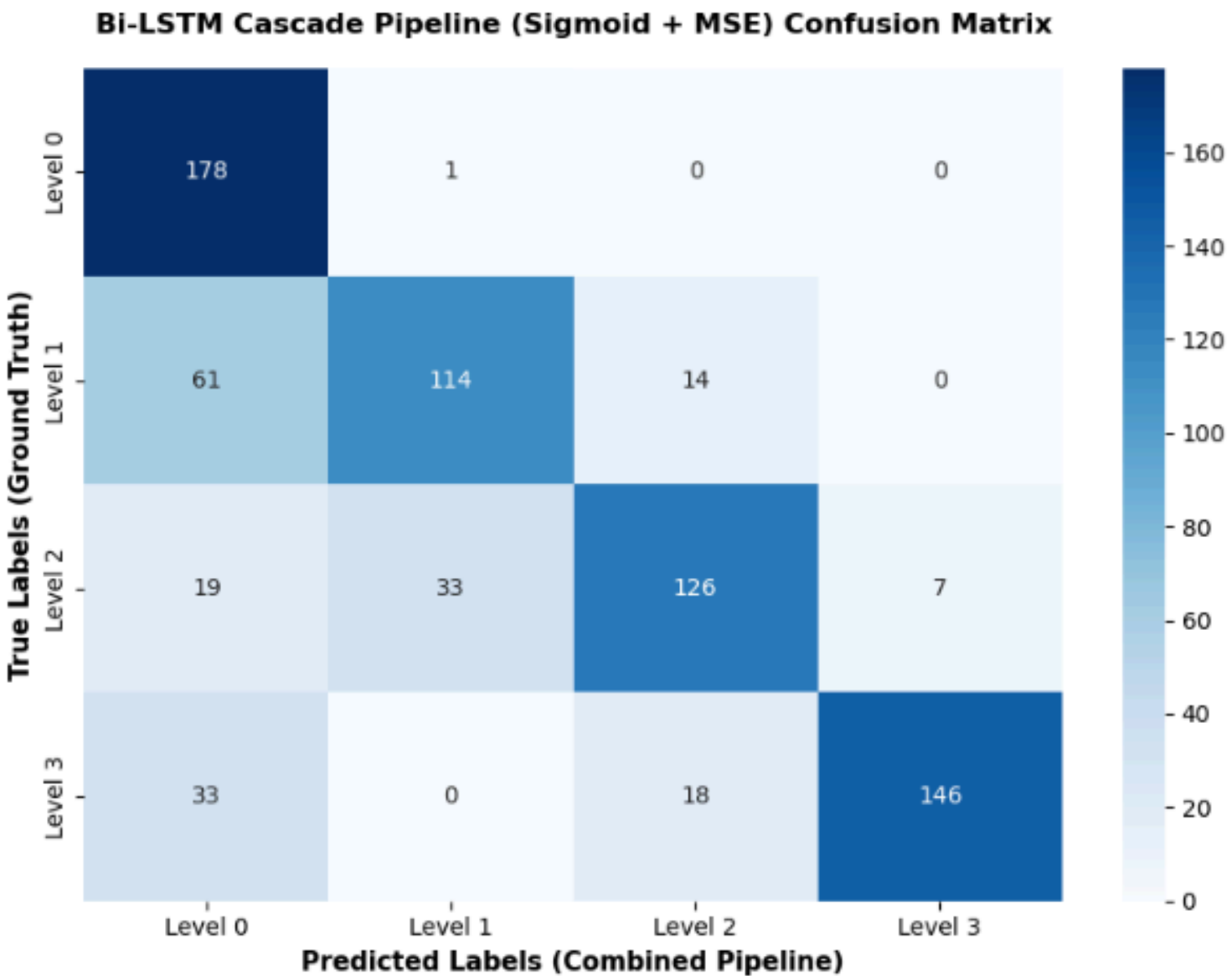
📄 Detailed Stage 2 Report:

	precision	recall	f1-score	support
Level 1	0.77	0.83	0.80	189
Level 2	0.70	0.71	0.71	185
Level 3	0.96	0.88	0.92	197
accuracy			0.81	571
macro avg	0.81	0.81	0.81	571
weighted avg	0.81	0.81	0.81	571

Stage 2: LSTM Severity Regressor Loss (MSE)



COMPLEX - BI LSTM OVERALL



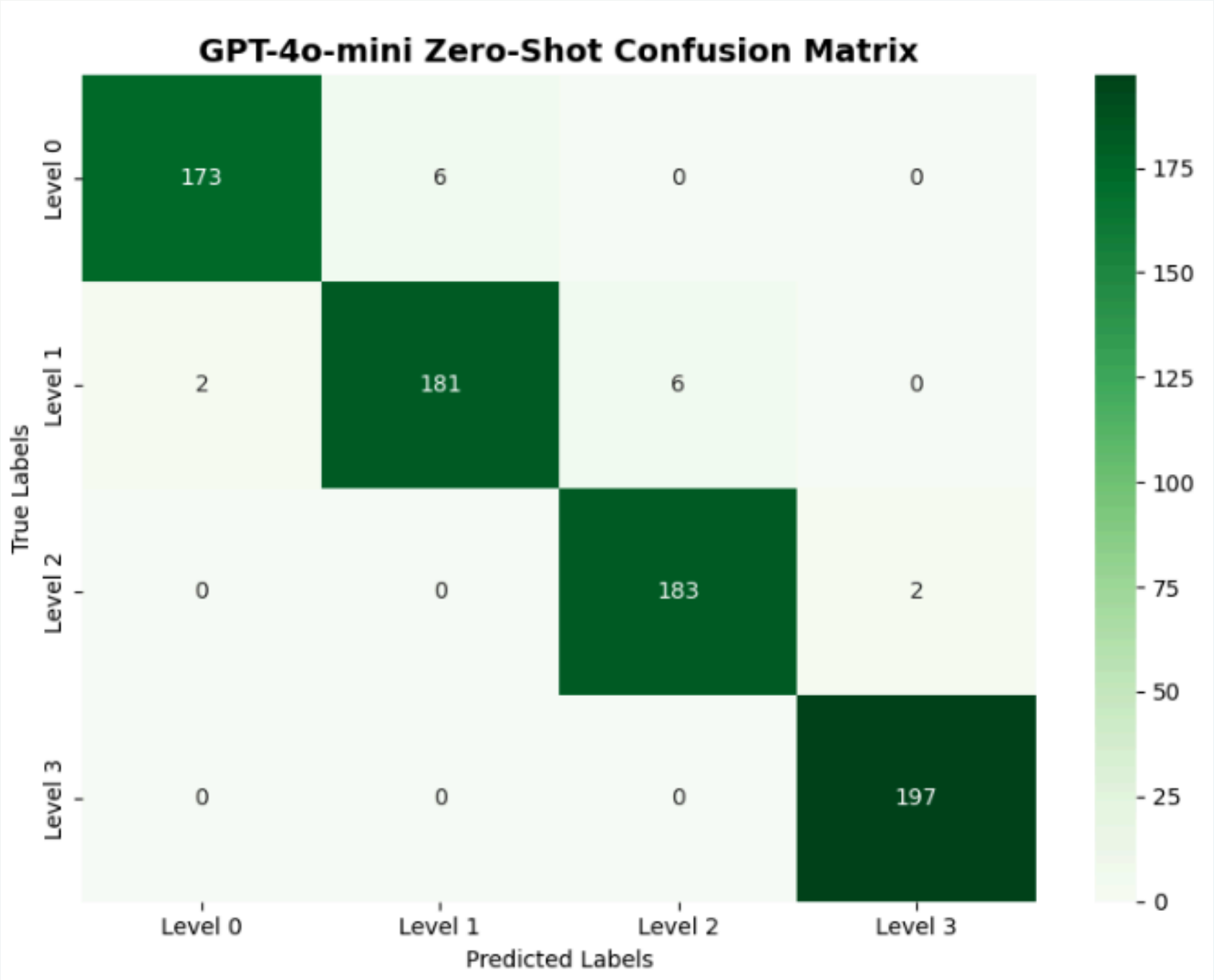
👑 **FINAL COMBINED CASCADE PIPELINE (Full Model 0-3):**

- Global Accuracy: 0.8360
- Global Macro F1-Score: 0.8391

📄 **Final Full Pipeline Classification Report:**

	precision	recall	f1-score	support
Level 0	0.99	0.93	0.96	179
Level 1	0.76	0.83	0.79	189
Level 2	0.68	0.71	0.70	185
Level 3	0.95	0.88	0.91	197
accuracy			0.84	750
macro avg	0.84	0.84	0.84	750
weighted avg	0.84	0.84	0.84	750

ZERO-SHOT GPT-4 - PROMPT BASED



🔥 Final GPT-4o-mini Zero-Shot Classification Report:

	precision	recall	f1-score	support
Level 0	0.99	0.97	0.98	179
Level 1	0.97	0.96	0.96	189
Level 2	0.97	0.99	0.98	185
Level 3	0.99	1.00	0.99	197
accuracy			0.98	750
macro avg	0.98	0.98	0.98	750
weighted avg	0.98	0.98	0.98	750

RESULTS: MODEL BENCHMARKING

Model	Part 1		Part 2		Overall	
	AUC	Macro F1	AUC	Macro F1	AUC	Macro F1
BERT	0.9787	0.9697	0.9947	0.9947	0.9747	0.974
BiLSTM	0.9813	0.9737	0.8074	0.807	0.836	0.8391
Zero-Shot GPT-4	-	-	-	-	0.98	0.98

RESULTS AND EXPLANATIONS

Specialized Encoder Bio_ClinicalBERT

Part 1

Robust handling of
noisy/informal text
(96.97% F1).

Part 2

Clinical Severity Mastery
Strong context-driven
regression (99.47% F1).

Low Error Variance

High stability tracking acute
risk boundaries (0.0092 MSE).

Neural Sequential Bi-LSTM

Part 1

Peak Binary Sorting
Exceptional at
administrative vs. clinical
separation (97.3% F1).

Part 2

Sequential Memory Decay
Recurrence struggles with
multi-level severity (80.7% F1).

Error Propagation

Final cascade pipeline accuracy
drops to 83.6%.

Generative LLM (GPT-4o mini)

Zero-Shot Domain Dominance

Maximum out-of-the-box
accuracy (98% F1) without
training.

End-to-End Mapping

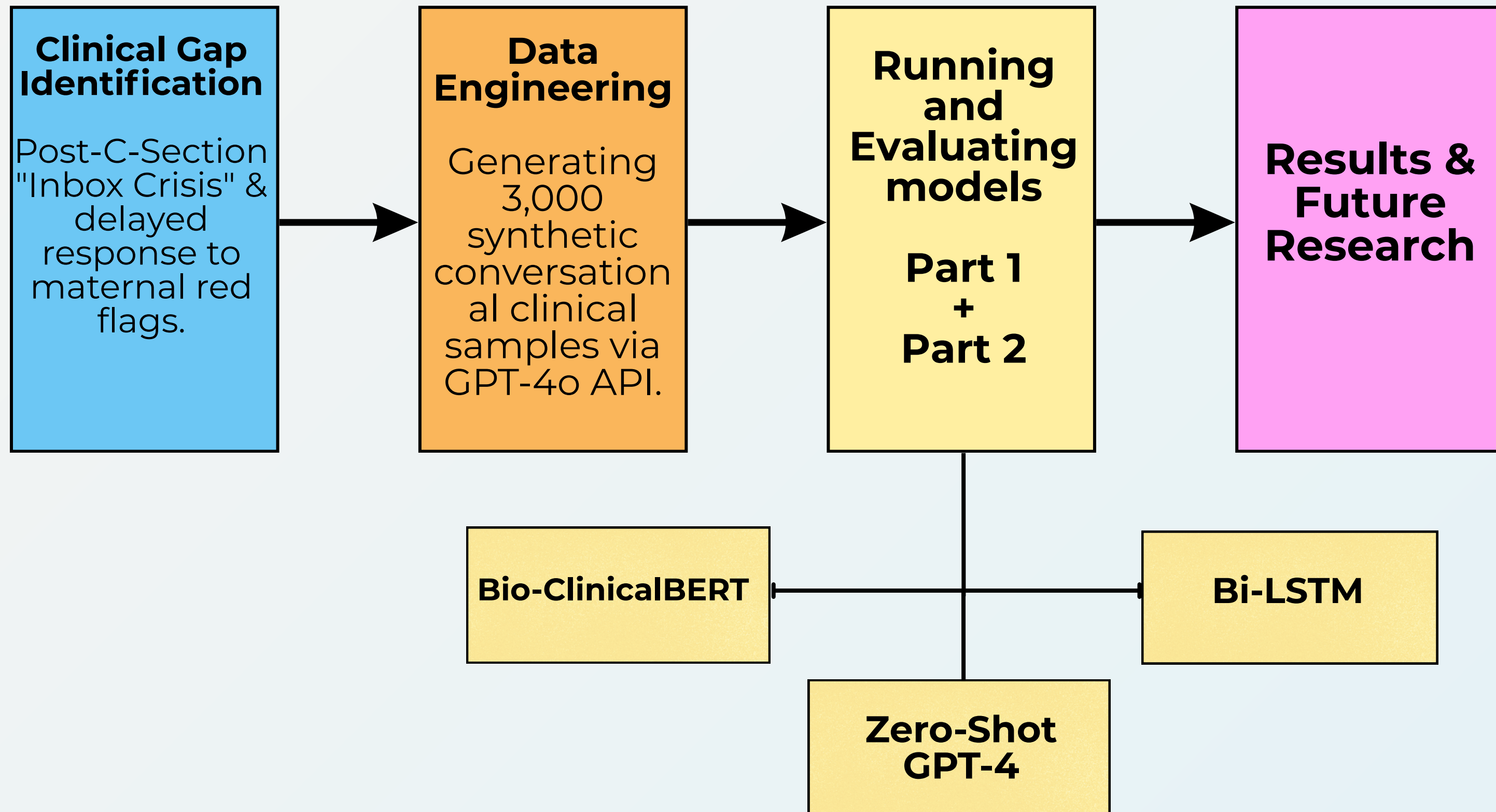
Direct classification without
complex multi-stage pipelines.

Infrastructure Dependencies

High robustness but
introduces cloud API privacy
trade-offs.

FROM CLINICAL GAP TO PREDICTIVE VERDICT

PROJECT ROADMAP



CONCLUSION & FUTURE RESEARCH

Healthcare Impact

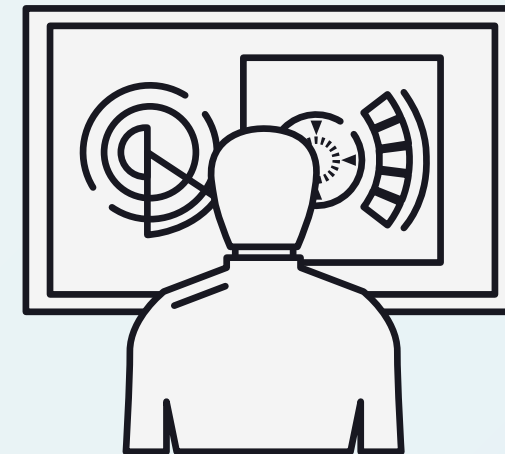
Automated, high-sensitivity triage solves the physician "Inbox Crisis" by instantly escalating critical postpartum warnings (Levels 2–3)



Core Takeaway

- BERT Baseline:
Outperformed the LSTM model, and also most realistic to use in the medical world

- GPT vs. LSTM :
Both have good results, GPT of course outperformed LSTM but is not realistic for use with medical data.



Future Scope

- Real-World Testing: Validating models using real, anonymous Israeli HMO messages.

Local & Global Deployment:
Adapting the framework for Israeli HMOs and international hospitals by integrating localized translation layers.

THANK YOU

